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4 / 4 submitted

Q1. Electric Battery Manufacturing

Score: 25.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

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int main() {
    int monthly, target = 5000;
    int annual, average;

    scanf("%d", &monthly);

    annual = monthly * 12;
    average = annual / 12;

    if (monthly > target) {
        printf("Yes Incentive\n");
    } else {
        printf("No Incentive\n");
    }

    if (monthly < (target * 80) / 100) {
        printf("Yes Inspection\n");
    } else {
        printf("No Inspection\n");
    }

    return 0;
}
```

INPUT

3500 42000 3500

OUTPUT

No Incentive
Yes Inspection

Q2. Maintenance

Score: 25.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    int load;
    char maintenance[10];
    float percentage;
    int available;

    scanf("%d%s", &load, maintenance);

    percentage = (load / 500.0) * 100;
    available = 500 - load;

    printf("Load Percentage = %.0f%%\n", percentage);
    printf("Available Capacity = %d MW\n", available);

    if(load > 500)
    {
        printf("Status = Grid Failure");
    }
    else if(strcmp(maintenance, "Yes") == 0)
    {
        printf("Under Maintenance");
    }
    else if(percentge ≤ 60)
    {
        printf("Status = Normal");
    }
    else if(percentge ≤ 85)
    {
        printf("Status = High Load");
    }
    else
    {
        printf("Status = Critical");
    }

    return 0;
}
```

INPUT

250 No

OUTPUT

Load Percentage = 50%
Available Capacity = 250 MW
Status = Normal

Q3. Medical

Score: 25.00 TC: 3/3 passed

CODE

```
#include<stdio.h>
#include<string.h>

int main(){
    int totalStock = 250000;
    int doses;

    scanf("%d",&doses);

    totalStock = totalStock-doses;

    if(doses ≥ 10000){
        printf("Hospital Status = Sufficient\n");
    }
    else if(doses ≥ 8000){
        printf("Hospital Status = Minimum Requirement\n");
    }
    else{
        printf("Hospital Status = Emergency Supply Required\n");
    }
    printf("Remaining Stock = %d doses\n",totalStock);

    return 0;
}
```

INPUT

8500

OUTPUT

Hospital Status = Minimum Requirement
Remaining Stock = 241500 doses

Q4. Storage

Score: 25.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
#include <string.h>

int main() {
    int used;
    char plan[20];
    int total, remaining, extraStorage;
    float extraCharge;

    scanf("%d %19s", &used, plan);

    if (strcmp(plan, "Basic") == 0) {
        total = 100;
    } else if (strcmp(plan, "Standard") == 0) {
        total = 500;
    } else if (strcmp(plan, "Enterprise") == 0) {
        total = 2000;
    } else {
        printf("Invalid Plan");
        return 0;
    }

    if (used ≤ total) {
        remaining = total - used;
        extraStorage = 0;
        extraCharge = 0;
    } else {
        remaining = 0;
        extraStorage = used - total;
        extraCharge = extraStorage * 4;
    }

    if (used ≤ total && (used*100)/total < 95) {

        printf("Total Storage = %d GB\n", total);
        printf("Remaining = %d GB\n", remaining);
        printf("Extra Storage = %d GB\n", extraStorage);
        printf("Extra Charge = Rs. %.0f\n", extraCharge);
        printf("Status = Normal\n");
    }

    else if (used > total) {
        printf("Remaining = %d GB\n", remaining);
        printf("Extra Storage = %d GB\n", extraStorage);
        printf("Extra Charge = Rs. %.0f\n", extraCharge);
        printf("Status = Extra Storage Used\n");
    }

    else {
        printf("Remaining = %d GB\n", remaining);
        printf("Extra Charge = Rs. %.0f\n", extraCharge);
        printf("Warning Generated\n");
        printf("Status = Storage Nearly Full\n");
    }

    return 0;
}
```

INPUT

1950 Enterprise

OUTPUT

Remaining = 50 GB
Extra Charge = Rs. 0
Warning Generated
Status = Storage Nearly Full